

Kunal Lohtia

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C++, C, SQL, HTML/CSS

Frameworks/Libraries: React, React Native, Node.js, Next.js, Express.js, Svelte, Redux, Flask, Tailwind CSS

AI/ML: LangChain, OpenAI API, Pandas, NumPy, Scikit-learn

Cloud & Databases: AWS, GCP, Docker, PostgreSQL, MySQL, MongoDB, Firebase, Supabase, Pinecone, Weaviate

Developer Tools: Git, GitHub Actions, Jira, Postman, Linux/Bash

EXPERIENCE

Software Engineer

Jan 2025 – Present

FloraPulse

Davis, CA

- Built a real-time irrigation web app (React, Node/Express, Supabase) ingesting telemetry from 700+ IoT sensors, with an automated grading engine that scores stem water potential to flag plant water stress for growers
- Implemented multi-layer caching (in-memory and local storage with API fallback), reducing average dashboard load time by 60% for users on low-bandwidth rural connections
- Designed a REST API for client data access, implementing SHA-256 hashed key authentication, role-based access control, rate limiting, and HMAC-SHA256 signed webhooks
- Developed an OpenAI and Pinecone-powered RAG chatbot, semantically indexing sensor documentation to handle 500+ monthly field queries and cutting resolution time to under 20s

Software Engineer (AI)

Jul 2024 – Jan 2025

CourseAssist

Davis, CA

- Engineered a multimodal RAG pipeline that processes video, audio, and diagrams using LangChain and OpenAI Embeddings with custom chunking and retrieval strategies, improving AI tutor response accuracy by 35%
- Built a semantic question clustering pipeline using OpenAI Embeddings and cosine similarity, surfacing GPT-summarized clusters to an instructor dashboard and reducing redundant student responses by 60%
- Integrated a subscription payment service with Stripe webhooks, implementing signature verification, event idempotency, and per-course rate limiting with Firestore

Technical Product Manager

Jan 2023 – Jan 2024

AggieWorks

Davis, CA

- Led system architecture design for a student marketplace app serving UC Davis, defining database schema, core service flows (listings, search/filtering, buyer-seller messaging), and technical requirements for an 8-engineer team
- Managed sprint planning and bug triage across a 60+ issue backlog, defining acceptance criteria and coordinating QA workflows that reduced user-reported bugs by 24% post-launch

Software Engineer Intern

Jan 2022 – Jan 2023

Xymob, Inc.

San Rafael, CA

- Shipped roster management and team coordination features in React Native with Redux and Firestore real-time listeners to a platform with 300K+ registered users across 35K+ teams
- Implemented user authentication with Firebase Auth and OAuth 2.0 social login, building role-based access control with distinct permission sets for coaches, parents, and players
- Developed a real-time messaging system with Firestore listeners, implementing typing indicators, read receipts, and offline message persistence

PROJECTS

Breast Cancer Tumor Predictor | *Python, Flask, Pandas, NumPy, Scikit-learn*

- Trained and compared Logistic Regression and Neural Network models (ReLU, Sigmoid) using 10-fold cross-validation and hyperparameter tuning, achieving 98% classification accuracy
- Deployed the models through a Flask web application, allowing users to input tumor attributes and receive real-time predictions

EDUCATION

University of California, Davis

B.S. Computer Science | Minor in Technology Management

Davis, CA

Dec 2023